

CPA Security and Semantic security

- Homework 1
 - Due Date: September 19, 2025
 - Resubmission: One week after grading

Today's Goal

- No deterministic encryption can be secure under the definition of multiple encryptions.
- CPA-Security. Any private-key encryption scheme that is CPA-secure, is also CPA-secure for multiple encryptions
 - First Hybrid Proof
- Semantic Security. EAV-security is equivalent to semantic security

Last time: Computational security

Two relaxations:

1. Security only against **Probabilistic Polynomial Time** adversaries.
2. Security can fail with a **negligible probability**
3. PRG security $\rightarrow$ EAV security

Computation indistinguishability against eavesdropper

Experiment $\text{PrivK}_{A,\Pi}^{\text{eav}}(n)$:

1. Adversary A is given input 1^n , and outputs two messages $m_0, m_1 \in \mathcal{M}$
s.t. $|m_0| = |m_1|$
2. A key k is generated using $\text{Gen}(1^n)$, a random bit $b \in \{0, 1\}$ is chosen, and $\text{Enc}(m_b)$ is given to A .
3. A outputs b'
4. Experiment outputs 1 if $b' = b$, else 0.

Definition (EAV security):
A scheme is **probabilistic poly** secure, if for all n

$$\begin{aligned} \Pr[\text{PrivK}_{A,\Pi}^{\text{eav}}(n) = 1] &= \frac{1}{2} \{ \Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{eav}}(n, 0)) = 0] + \Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{eav}}(n, 1)) = 1] \} \\ &= \frac{1}{2} \{ 1 - \Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{eav}}(n, 0)) = 1] + \Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{eav}}(n, 1)) = 1] \} \\ &= \frac{1}{2} - \frac{1}{2} \{ \Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{eav}}(n, 0)) = 1] - \Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{eav}}(n, 1)) = 1] \} \\ \Pr[\text{PrivK}_{A,\Pi}^{\text{eav}}(n) = 1] - \frac{1}{2} &= \frac{1}{2} \{ \Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{eav}}(n, 0)) = 1] - \Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{eav}}(n, 1)) = 1] \} \\ \left| \Pr[\text{PrivK}_{A,\Pi}^{\text{eav}}(n) = 1] - \frac{1}{2} \right| &= \frac{1}{2} |\Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{eav}}(n, 0)) = 1] - \Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{eav}}(n, 1)) = 1]| \end{aligned}$$

secure, if for all ϵ there exists a $\text{negl}(n)$ such that for

Or equivalently

$$|\Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{CPA}}(c = m_0)) = 1] - \Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{CPA}}(c = m_1)) = 1]| \leq \text{negl}(n)$$

- Suppose there is A breaks the EAV security

$$\Pr[\text{PrivK}_{A,\Pi}^{\text{eav}}(n) = 1] > \frac{1}{2} + \text{negl}(n)$$

- Distinguisher D with input 1^n and a string w of $(n+s)$ bits
 1. Run A to get $m_0, m_1 \in \mathcal{M}$ s.t. $|m_0| = |m_1| = n + s$
 2. Choose a random bit $b \in \{0,1\}$, set $c = w \oplus m_b$
 3. Give c to A and obtain output b' . Output 1 if $b'=b$, 0 otherwise

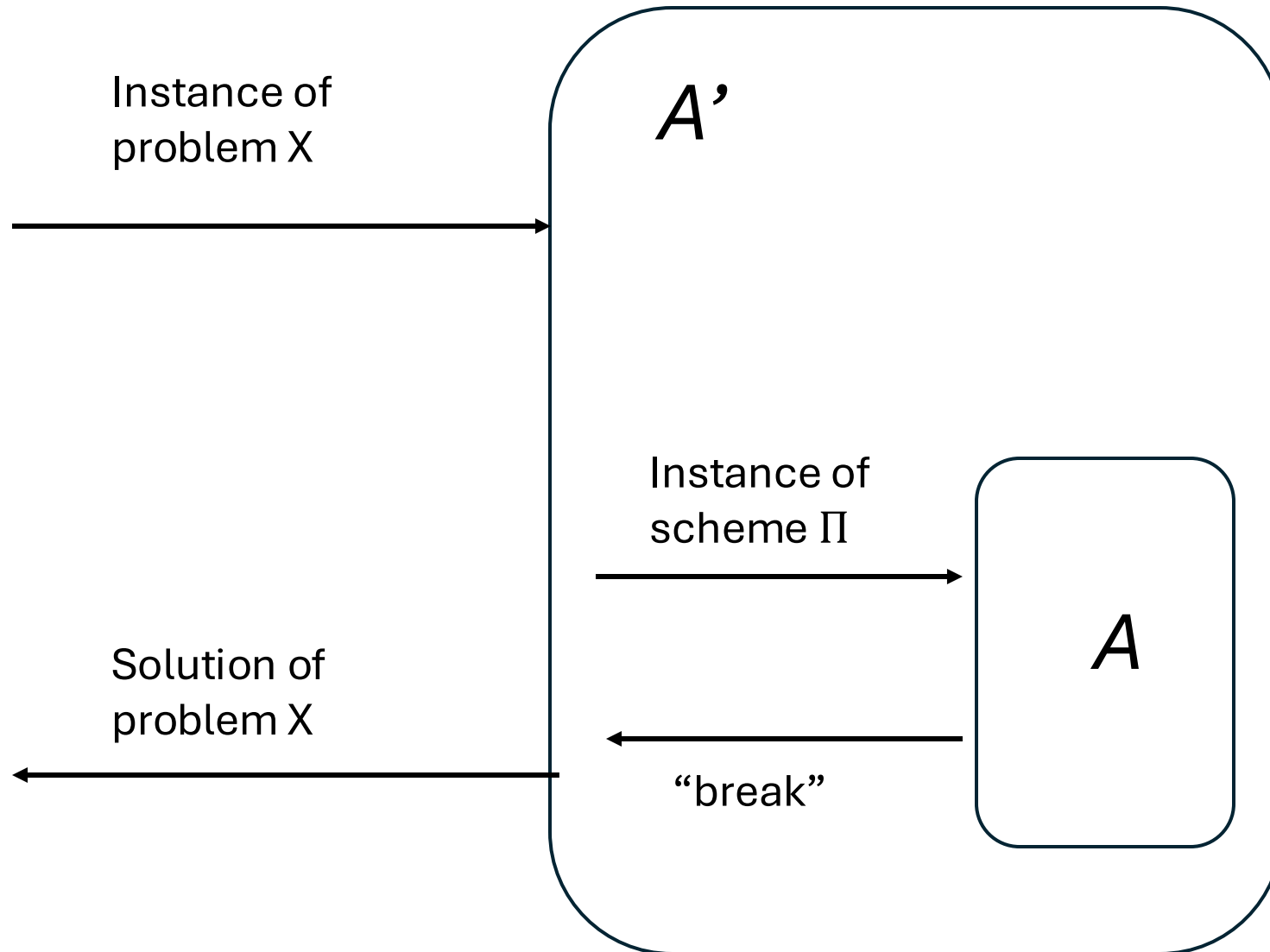
If w is a random string,

$$\Pr[D(w) = 1] = \Pr[\text{PrivK}_{A,\text{OTP}}^{\text{eav}}(n + s) = 1] = \frac{1}{2}$$

If $w = G(k)$,

$$\Pr[D(w) = 1] = \Pr[\text{PrivK}_{A,\Pi}^{\text{eav}}(n) = 1] > \frac{1}{2} + \text{negl}(n)$$

Proof by reduction



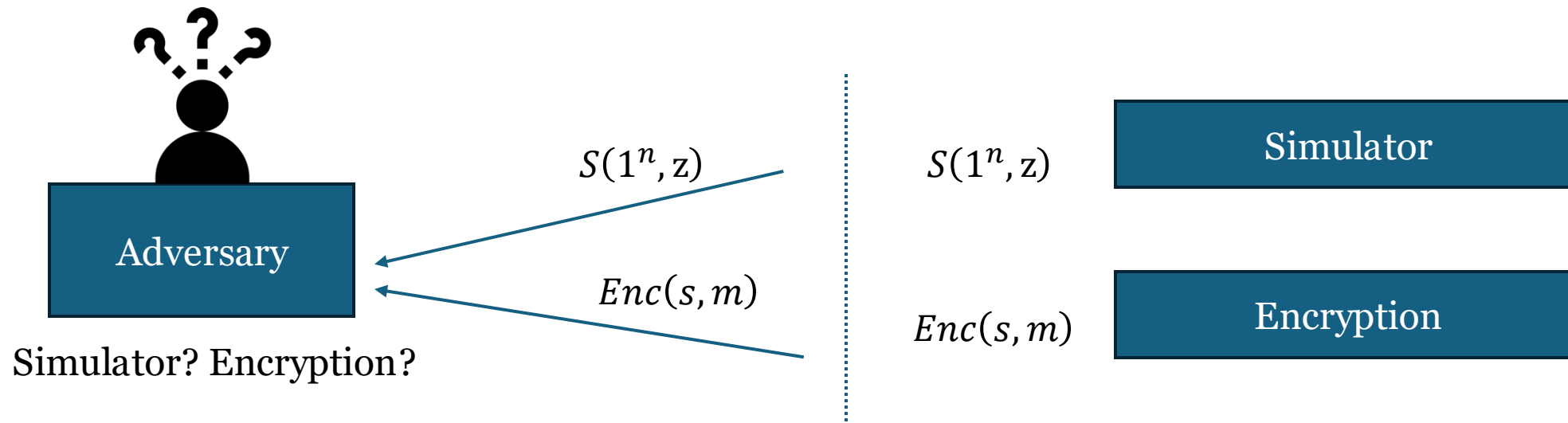
- X is the hard problem in the assumption
- Polynomial time reduction
- Non-negligible success probability

Proof

Semantic security

A secret-key encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ is semantically secure if there **exists** a PPT **simulator algorithm** S s.t. the following two experiments generate computationally indistinguishable outputs:

$$\left\{ \begin{array}{l} (m, z) \leftarrow M(1^n) \\ k \leftarrow \text{Gen}(1^n) \\ \text{Output } (\text{Enc}(s, m)) \end{array} \right\} \sim \left\{ \begin{array}{l} (m, z) \leftarrow M(1^n) \\ \text{Output } S(1^n, z) \end{array} \right\}$$



Semantic security

Theorem: Let $\Pi = (Enc, Dec)$ be a fixed-length private-key encryption scheme for messages of length ℓ that is EVA-secure. Then for all ppt. adversaries A and any $i \in \{1, \dots, \ell\}$, there is a negligible function $negl$ such that

$$\Pr[A(1^n, Enc_k(m)) = m^i] \leq \frac{1}{2} + negl(n)$$

the probability is taken over uniform $m \in \{0, 1\}^\ell$ and $k \in \{0, 1\}^n$, the randomness of A , and the randomness of Enc .

Proof.

In fact, adversary can't infer any information $f(m)$ about the message m .

Semantic security

- EAV-security is equivalent to semantic security.
- Proof
 - $SS \Rightarrow \text{EAV-security}$
 - $\text{EAV-security} \Rightarrow SS$

Security now?

The construction is not secure under the new definition

$$M_0 = \{\mathbf{0}^{n+s}, \mathbf{0}^{n+s}\}, M_1 = \{\mathbf{0}^{n+s}, \mathbf{1}^{n+s}\},$$

Necessity of probabilistic encryption

If Π is a (stateless) encryption scheme in which Enc is a deterministic function of the key and the message, then Π cannot have EVA-security under multiple encryptions.

Why?

Probabilistic encryption

Encrypting the same message multiple times gives **different** ciphertexts.

$r, c \leftarrow \text{Enc}_k(r, m)$, a random r is chosen for every encryption, and sent to the receiver

$m \leftarrow \text{Dec}_k(r, c)$

Chosen-plaintext attack (CPA) security

Experiment $\text{PrivK}_{A,\Pi}^{\text{CPA}}(n)$:

1. A key k is generated using $\text{Gen}(1^n)$
2. Adversary A is given input 1^n , and **oracle access to $\text{Enc}_k(\cdot)$** . A outputs two messages $m_0, m_1 \in \mathcal{M}$ s.t. $|m_0| = |m_1|$
3. A random bit $b \in \{0,1\}$ is chosen. $c = \text{Enc}_k(m_b)$ is given to A .
4. A continues to have **oracle access to $\text{Enc}_k(\cdot)$** and outputs b'
5. Experiment outputs 1 if $b=b'$, and 0 otherwise.

Definition: a private-key encryption $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ is CPA-secure, if for all probabilistic polynomial time adversaries A , there is a negligible function negl such that for all n ,

$$\Pr[\text{PrivK}_{A,\Pi}^{\text{CPA}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$$

or equivalently,

$$|\Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{CPA}}(c = m_0)) = 1] - \Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{CPA}}(c = m_1)) = 1]| \leq \text{negl}(n)$$

CPA VS EVA

- CPA security is **strictly** stronger than EVA security

What about multiple encryptions now?

Theorem: any private-key encryption scheme that is CPA-secure, is also CPA-secure for **multiple** encryptions.

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Theorem: any private-key encryption scheme that is CPA-secure, is also CPA-secure for **multiple** encryptions.

Proof: $t = 2$, simplified hybrid proof

1. Adversary A is given input 1^n , and outputs two **lists** of messages $M_0 = \{m_{01}, m_{02}, \dots, m_{0t}\}$, $M_1 = \{m_{11}, m_{12}, \dots, m_{1t}\}$, s.t. $|m_{0i}| = |m_{1i}| \forall i = 1, \dots, t$
2. A key k is generated using $\text{Gen}(1^n)$, a random bit $b \in \{0,1\}$ is chosen. A list $C = \{c_1, c_2, \dots, c_t\}$ is given to A , where $c_i = \text{Enc}_k(m_{bi}) \forall i = 1, \dots, t$
3. A outputs b'
4. Experiment outputs 1 if $b=b'$, and 0 otherwise.

A private-key encryption $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ is secure, if for all probabilistic polynomial time adversaries A , there is a negligible function negl such that for all n ,

$$|\Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{CPA}}(c = M_0)) = 1] - \Pr[\text{out}_A(\text{PrivK}_{A,\Pi}^{\text{CPA}}(c = M_1)) = 1]| \leq \text{negl}(n)$$

Proof